Lab-8 Ex5 - Decimal-Binary Conversion

Thomas is having difficulties calculating subtraction in binary number. Let's write a program to help him to double check the answer.

Write a program to perform the following tasks.

1. Ask user for 2 non-negative binary numbers, `b1` and `b2`. Assume `b1` is always greater than `b2`.
2. Convert them into decimal number `d1` and `d2`, respectively.
3. Subtract the decimal numbers (`diff_d = d1 - d2`).
4. Convert the difference from decimal back to binary.
5. Print out the result in both decimal and binary.

Some of the codes in the main function have been completed. Complete the remaining codes and also the two conversion functions.

You can assume the inputs are valid and each of the `b1`, `b2` and `diff_b` are non-negative and can be safely stored in an `int` variable. That means each of `b1`, `b2` and `diff_b` can be `0` to `1111111111` (binary) or `0` to `1023` (decimal).

User-input numbers are ***bold and italic*** in the following sample runs.

Sample Run #1

`b1 and b2?` ***110 100***

`The numbers (in decimal) are 6 and 4`

`The difference is 2 (decimal) or 10 (binary)`

Sample Run #2

`b1 and b2?` ***1010 1***

`The numbers (in decimal) are 10 and 1`

`The difference is 9 (decimal) or 1001 (binary)`

Sample Run #3

`b1 and b2?` ***1010110 0101111***

`The numbers (in decimal) are 86 and 47`

`The difference is 39 (decimal) or 100111 (binary)`